

Homework 25: Difference Equations & Differential Equations

BCS A, E, F, G

Problem.

a. *Difference Equation.*

For the given matrix

$$A = \begin{bmatrix} -4 & 2 \\ 3 & 1 \end{bmatrix},$$

and vector

$$x_0 = \begin{bmatrix} -1 \\ 4 \end{bmatrix}.$$

- i. Find the eigenvalues and corresponding eigenspaces.
- ii. Find $A^{10}x_0$.
- iii. Find $A^k x_0$. What happens as k becomes large, that is, $k \rightarrow \infty$? Give a reasonable explanation.
- iv. Find the solution of the difference equation $x_{k+1} = Ax_k$.

b. *System of Linear Differential Equations.*

Find the solution of the following system using eigenvalues and eigenvectors

$$\frac{dx_1}{dt} = -4x_1 + 2x_2, \quad \frac{dx_2}{dt} = 3x_1 + x_2,$$

subjected to initial conditions $x_1(0) = -1$ and $x_2(0) = 4$. Also find the steady state solution of the system.

c. *Recurrence Relation.*

Solve the recurrence relation

$$x_n = 4x_{n-1} - 3x_{n-2},$$

subjected to initial conditions $x_0 = 0, x_1 = 1$.

Submission Date: October 25, 2024 (Friday – 11:59PM)

Good Luck